09

Lecture 09

Evolved Packet Core (EPC)

هستهٔ بستهٔ تکامل‌یافته (EPC)

2G — 3G — **4G** — 5G — 6G

COURSE PART
Part II – 4G access and core (Lectures 5–13)

ESTIMATED STUDY TIME
7–9 hours (3.5 h reading, 4–5 h exercises)

DIFFICULTY
Intermediate

PREREQUISITES
Lectures 5–6; basic cryptography (symmetric keys, MAC); IP routing

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Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-09-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **Explain** the design goals of the EPC and the division of work among MME, HSS, S-GW, P-GW and PCRF.
- **LO2** **Describe** the EPS bearer model: default and dedicated bearers, TFTs and the QCI table, and **apply** GBR admission control.
- **LO3** **Trace** EPS-AKA and derive the EPS key hierarchy from K to the access-stratum keys.
- **LO4** **Trace** PDN connectivity and dedicated bearer activation, modification and deactivation.
- **LO5** **Describe** EMM/ECM states, tracking area updates and the EPC's role in handover.
- **LO6** **Analyse** the impact of node failures and **estimate** the signalling storm after a gateway restart.

2 Prerequisite knowledge

This lecture assumes the EPS architecture and reference points of Lecture 5 and the RRC/S1 procedures of Lecture 6. From security you need the idea of a shared symmetric key, of a message authentication code (a keyed checksum that proves integrity and origin) and of a key derivation function (KDF) that produces new keys from old ones without revealing them.

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
PDN connection	اتصال PDN	Association between a UE IP address and an APN, anchored in a P-GW.
Default bearer	حامل پیش‌فرض	First bearer of each PDN connection; always on.
Dedicated bearer	حامل اختصاصی	Additional bearer with its own QoS for selected flows.
TFT	قالب جریان ترافیک	Packet filters mapping IP flows to bearers.
QCI	شناسهٔ کلاس کیفیت خدمات	Index selecting priority, delay budget and loss rate.
EPS-AKA	توافق احراز هویت و کلید EPS	Mutual authentication producing K_{ASME}.
KDF	تابع استخراج کلید	HMAC-SHA-256-based derivation of new keys.
EMM / ECM	مدیریت تحرک / اتصال EPS	Registration state / signalling-connection state.
TAU	به‌روزرسانی ناحیهٔ ردیابی	UE procedure when entering a TA outside its list or periodically.
PCC	کنترل سیاست و صورت‌حساب	PCRF decisions enforced by the PCEF in the P-GW.

3 Introduction and motivation

The EPC was designed in the System Architecture Evolution (SAE) study with a clear brief: a *single*, *all-IP* core for LTE that could also serve legacy 3GPP access (via SGSN) and non-3GPP access (WLAN, CDMA2000), with lower latency, fewer nodes and a clean split between control and user planes [3]. The resulting core has only five essential node types, yet it must reproduce – with packets – everything the old CS and PS cores provided: subscriber authentication, mobility with session continuity, QoS for voice, charging and lawful interception.

This lecture is the system view of the EPC. It connects the pieces: how a subscriber is authenticated, how bearers are created and mapped to flows, how the UE's states are tracked, and what happens when a node fails. Lectures 10–13 then zoom into the protocols and nodes (Diameter, HSS, MME, gateways).

4 Fundamental concepts

4.1 Division of labour

- **MME** – control plane only: NAS signalling, authentication, security, tracking areas, paging, bearer control, gateway selection.
- **HSS** – subscription database and authentication centre.
- **S-GW** – user-plane mobility anchor and idle-mode buffer.
- **P-GW** – IP anchor towards each PDN, policy and charging enforcement (PCEF).
- **PCRF** – policy and charging decisions (PCC rules).

The MME and the gateways are connected by S11 (GTPv2-C), the gateways by S5/S8; the MME talks to the HSS over S6a and the P-GW to the PCRF over Gx, both Diameter.

4.2 The bearer model

Definition 1 • EPS bearer and TFT

An EPS bearer is the unit of QoS in LTE: all packets on a bearer receive the same forwarding treatment. A *traffic flow template* (TFT) is a set of packet filters (source/destination address and port ranges, protocol, type of service) that decides which bearer a packet uses: the *UL TFT* in the UE for uplink packets, the *DL TFT* in the P-GW for downlink packets.

Each PDN connection starts with a default bearer without a TFT (it catches everything). Dedicated bearers are created when a service needs different QoS – typically triggered by the PCRF when an application function (e.g. IMS) requests it. Every bearer is characterised by

- **QCI** – a standardised class with priority, packet delay budget (PDB) and packet error loss rate (PELR) (Table 1);
- **ARP** – allocation and retention priority, used for admission and pre-emption;
- for GBR bearers, **GBR** and **MBR** (guaranteed and maximum bit rate);
- for non-GBR bearers collectively, the **APN-AMBR** and **UE-AMBR** aggregate limits.

Table 1. Standardised QCI characteristics (TS 23.203, Table 6.1.7-A, QCI 1–9) [5].

QCI	Type	Priority	PDB [ms]	PELR	Example services
1	GBR	2	100	10^{-2}	Conversational voice (VoLTE)
2	GBR	4	150	10^{-3}	Conversational video (live)
3	GBR	3	50	10^{-3}	Real-time gaming
4	GBR	5	300	10^{-6}	Non-conversational (buffered) video
5	non-GBR	1	100	10^{-6}	IMS signalling
6	non-GBR	6	300	10^{-6}	Buffered video, TCP applications (premium)
7	non-GBR	7	100	10^{-3}	Voice, live video, interactive gaming
8	non-GBR	8	300	10^{-6}	TCP applications (e.g. web, e-mail)
9	non-GBR	9	300	10^{-6}	TCP applications, typical default bearer

4.3 EMM and ECM states

Two orthogonal state machines describe a UE in the EPC (Figure 1): *EPS mobility management* (EMM-DEREGISTERED/REGISTERED) says whether the network knows the UE; *EPS connection management* (ECM-IDLE/CONNECTED) says whether a NAS signalling connection (RRC + S1) exists. A registered UE in ECM-IDLE keeps its IP address and bearers in the core (S5/S8 tunnels stay up), but has no radio or S1-U resources; the S-GW buffers downlink data and asks the MME to page.

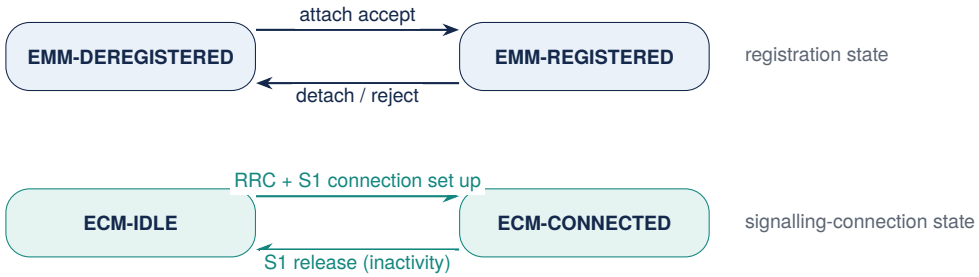


Figure 1. EMM (registration) and ECM (signalling connection) states of a UE in the EPC.

5 Detailed technical explanation

5.1 Authentication: EPS-AKA

EPS-AKA [2] extends UMTS AKA (Lecture 3). The HSS generates an *EPS authentication vector* (RAND, XRES, AUTN, K_{ASME}); the novelty is that K_{ASME} is bound to the serving network identity, so a vector issued for one network is useless in another. The MME sends RAND and AUTN; the USIM checks AUTN (which proves that the challenge comes from the home network and is fresh, via the sequence number SQN), computes RES and the same keys. The MME compares RES with XRES; both sides now share K_{ASME}, which never leaves the MME. The detailed Milenage computations are in Lecture 11.

5.2 NAS and AS security

Security is layered:

- **NAS security** (UE–MME): ciphering and integrity protection of NAS messages with K_{NASenc} and K_{NASint}, activated by the NAS Security Mode Command.
- **AS security** (UE–eNB): integrity and ciphering of RRC (K_{RRCint}, K_{RRCenc}) and ciphering of user data (K_{UPenc}) in PDCP, derived from K_{eNB} which the MME sends to the eNB in Initial Context Setup.

User data is *not* integrity-protected in LTE (only ciphered) – a deliberate trade-off for throughput that 5G revisits. The algorithms are EEA1/EIA1 (SNOW 3G), EEA2/EIA2 (AES) and EEA3/EIA3 (ZUC); EEA0/EIA0 are null algorithms, EIA0 being allowed only for emergency calls without authentication.

5.3 Session management

PDN connectivity. At attach (and later for additional APNs) the UE requests a PDN connection. The MME selects a P-GW from the APN (DNS-based) and an S-GW from the UE's location, and sends Create Session Request; the P-GW allocates the IP address (IPv4, IPv6 prefix or both), asks the PCRF for policy (Gx CCR-I/CCA-I) and creates the default bearer.

Dedicated bearer activation. Normally network-initiated (Figure 5): the PCRF provisions PCC rules, the P-GW sends Create Bearer Request with TFT and QoS through the S-GW to the MME, which sets up the E-RAB in the eNB and sends the NAS Activate Dedicated EPS Bearer Context Request to the UE.

Modification and deactivation. QoS or TFT changes use Update Bearer Request; removal uses Delete Bearer Request. Deleting the default bearer deletes the whole PDN connection.

5.4 Mobility in the EPC

- **Tracking area update (TAU)**: when the UE enters a TA not in its list, or when the periodic timer T3412 expires; the MME may change (context transfer from the old MME) and so may the S-GW.
- **X2 handover**: the core only switches the path (Lecture 4).
- **S1 handover**: used without X2 or when the MME or S-GW changes; the source and target eNBs exchange handover messages via the MME(s).
- **Inter-RAT mobility**: to UTRAN/GERAN through the SGSN (S3/S4 interfaces); the S-GW remains the anchor, the P-GW keeps the IP address.

5.5 Failure scenarios

- **MME failure** – with S1-flex the eNB connects to other MMEs of the pool; UEs whose contexts were lost are recovered when they next contact the network (TAU or service request is rejected with an implicit detach, followed by re-attach).
- **S-GW failure** – detected through GTP echo and the *restart counter* carried in the Recovery information element. The MME and P-GW delete the affected PDN connections; idle UEs are not reachable until they re-attach or the MME restores the sessions on another S-GW (restoration procedures).
- **P-GW failure** – the IP anchor is lost: the UE's IP address and all its sessions on that APN disappear; applications must reconnect.
- **HSS unavailable** – already registered UEs continue (the MME caches subscription and unused vectors); new attaches of UEs unknown to the MME fail.

6 Mathematical models

6.1 GBR admission control

Problem. How many GBR bearers can a cell or gateway admit?

Model. A resource (cell, link) has an average capacity C; the operator reserves a fraction ρ_{GBR} for GBR traffic. Bearer i requests GBR_i. Admission requires

$$\sum_{i \in \mathcal{A}} \mathrm{GBR}_i + \mathrm{GBR}_{\mathrm{new}} \le \rho_{\mathrm{GBR}}\, C, \tag{1}$$

otherwise the request is rejected, or a bearer with lower ARP is pre-empted if the new one may pre-empt and the old one is pre-emptable.

Example. $C = 30\,\mathrm{Mbit/s}$ average cell throughput, $\rho_{\mathrm{GBR}} = 0.5$, VoLTE GBR $= 40\,\mathrm{kbit/s}$: at most $15 \times 10^6 / 40 \times 10^3 = 375$ calls.

Limits. In a radio cell C is not constant: it depends on where the users are (their MCS). Radio admission therefore estimates the *resource* (RBs) a bearer will need at its current channel quality rather than its bit rate (Exercise 3).

6.2 The EPS key hierarchy

Model. Keys are derived with the 3GPP KDF, HMAC-SHA-256 keyed with the parent key over a string S that encodes the purpose (an FC byte and parameters) [2]:

$$K_{\mathrm{ASME}} = \mathrm{KDF}\big(\mathrm{CK} \,\|\, \mathrm{IK},\ \mathrm{SN\ id},\ \mathrm{SQN} \oplus \mathrm{AK}\big), \tag{2a}$$

$$K_{\mathrm{eNB}} = \mathrm{KDF}\big(K_{\mathrm{ASME}},\ \text{uplink NAS COUNT}\big), \tag{2b}$$

$$K_{\mathrm{NASint}} = \mathrm{KDF}\big(K_{\mathrm{ASME}},\ \text{alg. type},\ \text{alg. id}\big)\ \text{(truncated to 128 bits)}, \tag{2c}$$

and analogously for K_{NASenc}, K_{RRCint}, K_{RRCenc} and K_{UPenc} from K_{eNB} (Figure 3).

Interpretation. Each key is bound to its context: the serving network (Eq. (2a)), the NAS COUNT, hence a fresh K_{eNB} at every transition to connected mode (Eq. (2b)), and the algorithm. Compromise of one eNB's keys reveals nothing about K_{ASME} (one-way KDF). At handover a new K^*_{eNB} is derived so the target does not learn the source's keys.

Brute force. A 128-bit algorithm key has $2^{128} \approx 3.4 \times 10^{38}$ values; at 10^{12} trials per second an exhaustive search takes about 1.1×10^{19} years. Security therefore rests on protecting K and on the algorithms, not on key length.

6.3 Signalling storm after a gateway restart

Problem. An S-GW serving N UEs restarts; all of them must re-establish their sessions. How long does recovery take, and what load does it put on the MME?

Model. Each recovery costs m messages at the MME; the MME processes at most μ messages per second. If all UEs try immediately, the MME is saturated for

$$T_{\mathrm{rec}} = \frac{N\,m}{\mu}. \tag{3}$$

If recovery attempts are spread uniformly over a window $T_{\mathrm{w}} > T_{\mathrm{rec}}$, the offered signalling rate is Nm/T_{w} and the MME stays below saturation.

Example. $N = 10^6$, $m = 10$, $\mu = 2 \times 10^4$ msg/s: $T_{\mathrm{rec}} = 500\,\mathrm{s}$. Spreading over $T_{\mathrm{w}} = 900\,\mathrm{s}$ gives 1.1×10^4 msg/s (56% of capacity) instead of an instantaneous burst of 10^7 messages, which would overflow queues, time out, and trigger retries that make the storm worse.

Lesson. Restoration procedures and randomised timers exist to turn a burst into a spread load – the same positive-feedback hazard as RACH overload (Lecture 8), one level higher.

7 Network architecture and diagrams

Figure 2 shows the bearer as seen end to end. The radio bearer, S1 bearer and S5/S8 bearer are bound one-to-one; the S-GW and eNB simply relay packets between tunnels, while the TFTs at the two ends decide the bearer. The E-RAB (radio + S1) is the part that is released in ECM-IDLE; the S5/S8 part persists.

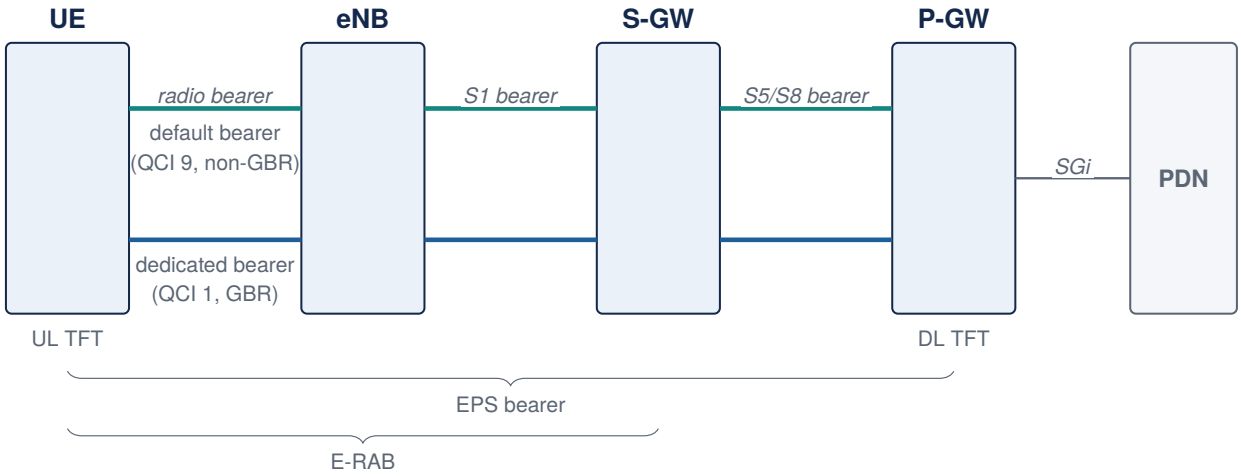


Figure 2. EPS bearer structure with a default and a dedicated bearer. The UL TFT in the UE and the DL TFT in the P-GW map IP flows to bearers.

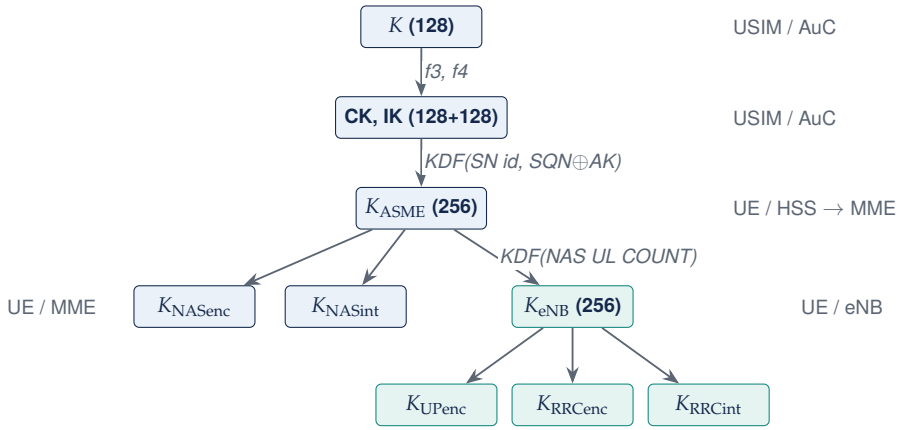


Figure 3. EPS key hierarchy. Key lengths in bits; algorithm keys are truncated to 128 bits.

8 Step-by-step procedures

Procedure 1 • EPS-AKA and NAS security activation

1. The UE sends Attach Request with its GUTI (or IMSI if it has none).
2. If no vector is available, the MME sends S6a Authentication Information Request with the IMSI and the serving network identity; the HSS returns one or more EPS vectors.
3. The MME sends Authentication Request (RAND, AUTN, key set identifier).
4. The USIM verifies AUTN (MAC and SQN freshness), computes RES, CK, IK; the UE derives K_{ASME} and returns Authentication Response (RES).
5. The MME checks RES = XRES and sends Security Mode Command (selected algorithms, integrity protected with K_{NASint}); the UE verifies it and answers Security Mode Complete, ciphered and integrity-protected.

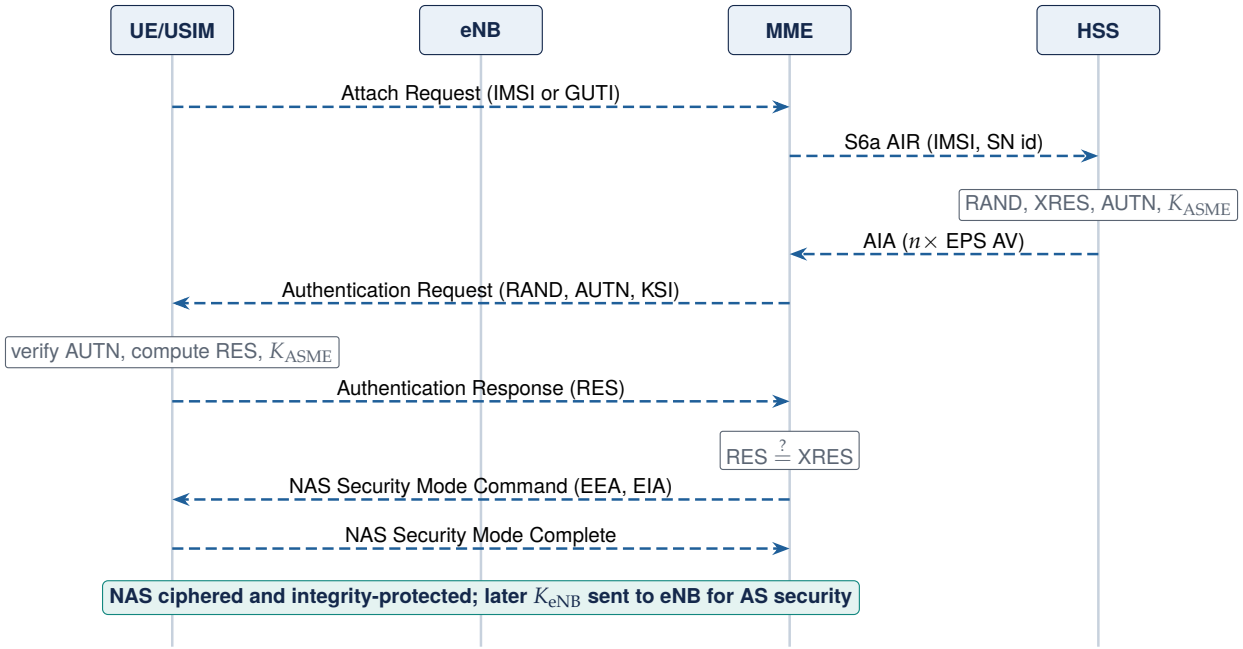


Figure 4. EPS-AKA and NAS security mode (Procedure 1).

Procedure 2 • Network-initiated dedicated bearer activation

1. An application function (e.g. the IMS P-CSCF) requests resources over Rx; the PCRF decides PCC rules and pushes them to the P-GW over Gx.
2. The P-GW binds the rule to a new bearer (QCI, ARP, GBR/MBR, DL TFT) and sends GTPv2-C Create Bearer Request to the S-GW, which forwards it to the MME.
3. The MME sends S1AP E-RAB Setup Request carrying the NAS Activate Dedicated EPS Bearer Context Request (UL TFT, QoS).
4. The eNB admits the bearer, configures a radio bearer with RRC Connection Reconfiguration and answers E-RAB Setup Response; the UE answers the NAS request with *Accept*.
5. Create Bearer Response travels back to the P-GW, which confirms to the PCRF (Gx RAA). Packets matching the TFTs now use the new bearer.

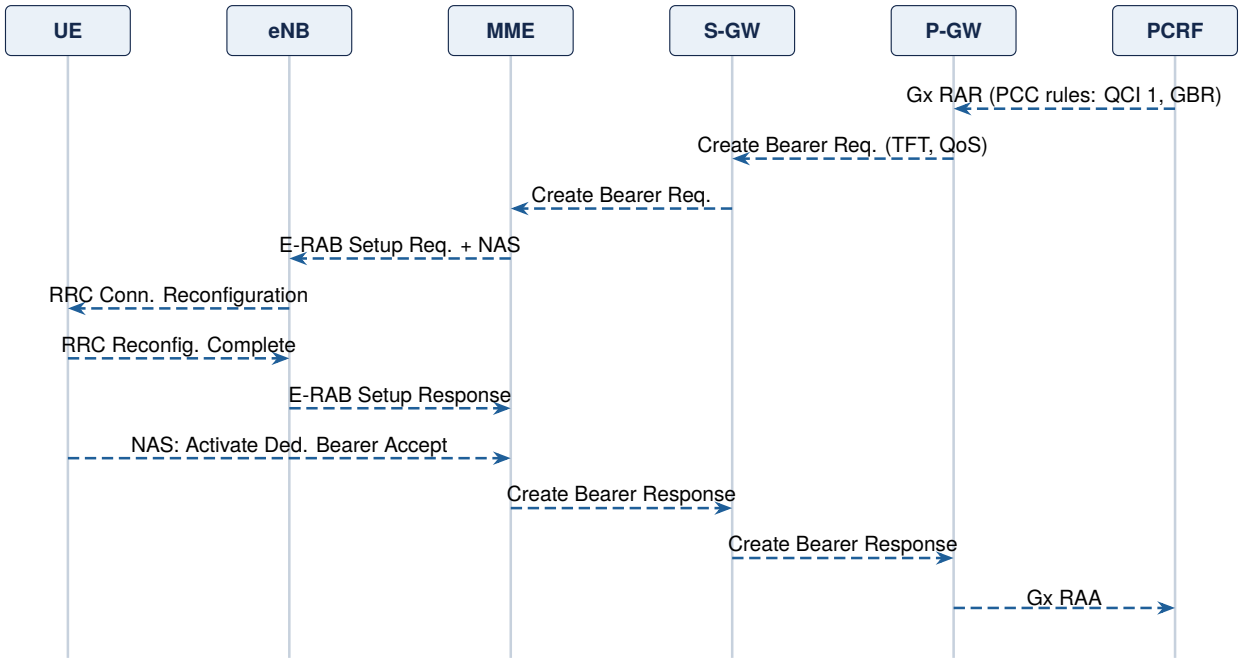


Figure 5. Dedicated bearer activation triggered by policy (Procedure 2).

9 Worked examples

Worked example 1 • Which bearer carries the packet?

Problem. A UE has a default bearer (QCI 9) and a dedicated bearer (QCI 1) whose DL TFT matches UDP packets from 10.1.2.3, ports 40000–40100. Which bearer carries (a) an RTP packet from 10.1.2.3:40050, (b) a TCP packet from 10.1.2.3:443, (c) a DNS response?
Solution. (a) Dedicated bearer (all filter components match). (b) Default bearer (protocol and port do not match). (c) Default bearer (no filter matches; the default bearer has no TFT and receives the rest).
Interpretation. Bearer selection is pure packet classification in the P-GW (downlink) and UE (uplink); the radio then applies the QCI's treatment to the bearer.

Worked example 2 • Admission by radio resources

Problem. A VoLTE bearer needs 40 kbit/s. At its MCS a cell-centre UE carries 800 bits per RB pair (one RB pair for one 1-ms subframe); a cell-edge UE only 160 bits. If 25 RB pairs per subframe are reserved for GBR traffic, how many calls of each kind can be admitted?
Solution. RB pairs per subframe needed $=$ GBR$/$(bits per RB pair $\times$ 1000 s^{-1}): centre $40\,000/(800 \times 1000) = 0.05$, edge $40\,000/(160 \times 1000) = 0.25$. Admissible: $25/0.05 = 500$ centre users or $25/0.25 = 100$ edge users.
Interpretation. The same GBR costs five times more radio resource at the cell edge – admission must be resource-based (Section 6.1).

Worked example 3 • Key freshness

Problem. A UE goes from idle to connected 200 times per day with the same K_{ASME}. How many different K_{eNB} values does it use, and why does that matter?
Solution. 200, because each transition uses a new uplink NAS COUNT in Eq. (2b). Even if the same eNB serves all connections, PDCP COUNT values restart with each new key, so keystreams are never reused under one key.

10 Technology comparison

Table 2. Packet core functions: GPRS/UMTS PS core versus EPC versus 5GC.

	GPRS/UMTS PS	EPC	5GC
Mobility/session control	SGSN (both)	MME (mobility), S-/P-GW-C (session)	AMF (mobility), SMF (session)
User-plane anchor	GGSN	P-GW (S-GW as mobility anchor)	UPF (one or more)
QoS unit	PDP context	EPS bearer (QCI)	QoS flow (5QI) within a PDU session
IP at registration	No (separate activation)	Yes (default bearer)	Optional (PDU session separate)
Authentication	UMTS AKA (quintets)	EPS-AKA (K_{ASME})	5G-AKA / EAP-AKA′, home control
Subscriber database	HLR (MAP)	HSS (Diameter)	UDM/UDR (HTTP/2)

Trade-offs. The bearer model gives precise, per-bearer QoS but every QoS change needs a signalling procedure through four nodes; 5G replaces it with QoS flows that can be added inside an existing session with less signalling (Lecture 22). The MME's concentration of NAS, security and bearer control is efficient but makes it a large stateful node; 5G splits it into AMF and SMF.

11 Key takeaways

Key takeaways

- EPC: MME (control), HSS (subscribers), S-GW (mobility anchor), P-GW (IP anchor, PCEF), PCRF (policy).
- QoS unit = EPS bearer (QCI, ARP, GBR/MBR); TFTs map flows to bearers; default bearer always on.
- EPS-AKA binds K_{ASME} to the serving network; KDF chain $K \rightarrow$ CK,IK $\rightarrow K_{\text{ASME}} \rightarrow K_{\text{eNB}} \rightarrow$ AS keys.
- NAS security UE–MME, AS security UE–eNB; user data ciphered but not integrity-protected in LTE.

- EMM (registered?) and ECM (signalling connection?) are independent state machines.
- GBR admission must be resource-based; a gateway restart can cause a signalling storm of Nm/μ seconds unless recovery is spread.

12 Exercises

Exercise 1 • Level 1 – conceptual • Idle UE

List which resources exist for a UE in EMM-REGISTERED + ECM-IDLE: IP address, S5/S8 tunnel, S1-U tunnel, radio bearer, eNB context, MME context. What triggers their re-creation?

Exercise 2 • Level 1 – conceptual • Serving-network binding

Explain what attack is prevented by including the serving network identity in the derivation of K_{ASME}.

Exercise 3 • Level 2 – numerical • Admission

Repeat Example 2 for a mix of 70% centre and 30% edge users. How many VoLTE calls can be admitted?

Exercise 4 • Level 2 – numerical • Periodic TAU load

An MME serves 5×10^6 registered UEs with T3412 $= 54\,\text{min}$. Each periodic TAU costs 6 messages. Compute the average message rate. What happens to the load if T3412 is reduced to 12 min?

Exercise 5 • Level 3 – analytical • Storm with retries

In Section 6.3, assume that requests waiting longer than 30 s time out and are retried once. Model the MME as a single queue with service rate μ and estimate the total number of messages processed when all N UEs try at once. Compare with spreading over $T_{\text{w}} = 900\,\text{s}$.

Exercise 6 • Level 3 – analytical • Bearer versus flow

A UE runs a VoLTE call, a video call and background web traffic. Count the GTP tunnels on S1-U and S5 in LTE (one PDN for IMS, one for the Internet) and discuss the signalling needed to add a second video stream. How would a flow-based model reduce it?

Solutions

Solution to Exercise 1

Kept: IP address, S5/S8 tunnels, MME context (and S-GW/P-GW contexts). Released: S1-U tunnel, radio bearers, eNB context. They are re-created by a Service Request – triggered by the UE (uplink data or signalling) or by paging after the S-GW receives downlink data and sends a Downlink Data Notification to the MME.

Solution to Exercise 2

Without binding, a vector (or key) obtained by one serving network could be used by another, e.g. a compromised or rogue visited network could reuse vectors to impersonate a different network to the UE, or a network could claim to be a cheaper one for billing. Because both the HSS and the UE derive K_{ASME} with the SN id, a mismatch between the claimed network and the network that requested the vector produces different keys and security mode fails.

Solution to Exercise 3

Mean cost per call $= 0.7 \times 0.05 + 0.3 \times 0.25 = 0.11$ RB pairs per ms; calls $= 25/0.11 \approx 227$.

Solution to Exercise 4

TAU rate $= 5 \times 10^6/(54 \times 60) = 1543$ per s; messages $= 6 \times 1543 = 9259$ msg/s. With 12 min the rate increases by $54/12 = 4.5\times$ to about 41 700 msg/s – shorter timers detect lost UEs faster but cost signalling and UE battery.

Solution to Exercise 5

Total work $Nm = 10^7$ messages at $\mu = 2 \times 10^4$ msg/s needs 500 s. Requests queued longer than 30 s time out: only the first $30\mu/m = 6 \times 10^4$ UEs complete in the first 30 s; the others retry, adding up to N new attempts. A crude bound: with one retry each, offered work approaches $2Nm$ and recovery time about 1000 s, with many UEs failing twice. Spreading over 900 s keeps the queue short (load 56%), no timeouts, total work exactly Nm – faster overall despite the deliberate delay.

Solution to Exercise 6

IMS PDN: default bearer (QCI 5), dedicated QCI 1 (voice) and QCI 2 (video): 3 bearers. Internet PDN: default bearer: 1. Total 4 EPS bearers, i.e. 4 S1-U tunnels and 4 S5 tunnels. A second video stream with the same QCI can be added to the existing QCI 2 bearer by updating its TFT (Update Bearer procedure, still through PCRF, P-GW, S-GW, MME, eNB and UE); a new QCI would need a new bearer. In 5G all flows of a PDU session share one N3 tunnel; adding a QoS flow is a rule update in SMF, UPF and gNB with less per-flow tunnel state (Lecture 22).

13 Further reading

EPC procedures are specified in [3], policy and charging control and the QCI table in [5], security in [2], NAS in [4] and GTPv2-C in [1]. Olsson et al. [6] give a highly readable system description of the EPC.

References

[1] 3GPP, *3GPP evolved packet system (EPS); evolved general packet radio service (GPRS) tunnelling protocol for control plane (GTPv2-C); stage 3*, TS 29.274, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/29274.htm.

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[3] 3GPP, *General packet radio service (GPRS) enhancements for evolved universal terrestrial radio access network (E-UTRAN) access*, TS 23.401, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/23401.htm.

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[6] M. Olsson, S. Sultana, S. Rommer, L. Frid, and C. Mulligan, *EPC and 4G Packet Networks: Driving the Mobile Broadband Revolution*, 2nd ed. Academic Press, 2013.